

Teorija polj
Vektorski operatorji

$$\begin{aligned} f &= \text{skalarno polje, } \mathbf{F} = \text{vektorsko polje} \\ \text{grad: } \nabla f &= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \\ \text{div: } \nabla \cdot \mathbf{F} &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \\ \text{rot: } \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = (Py - Qz, Pz - Rx, Qx - Py) \\ \text{Laplacijan: } \nabla^2 f &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \end{aligned}$$

Lastnosti polj

$$\begin{aligned} \text{Irotacijalno: } \nabla \times \mathbf{F} &= 0 \\ \text{Solenoidno: } \nabla \cdot \mathbf{F} &= 0 \\ \text{Potencial: } \vec{v}(x, y, z) &= \textit{gradu}(x, y, z) \\ \text{u = skalarno p. in potencial polja v} \end{aligned}$$

Prostorska geometrija
Koordinatni sistemi

$$\begin{aligned} \text{Polarni: } x &= r \cos \theta, \quad y = r \sin \theta \quad J = r \\ \text{Cilindrični: } x &= r \cos \theta, \quad y = r \sin \theta, \quad z = z \quad J = r \\ \text{Sferični: } x &= r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta, \\ z &= r \cos \phi \quad J = r^2 \sin \phi \end{aligned}$$

Dodatno

$$\begin{aligned} \text{Ločna dolžina: } s &= \int_a^b |\mathbf{r}'(t)| \, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} \, dt \\ \text{Tangentni vektor: } \mathbf{T} &= \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \\ \text{Normalna ravnina: } \mathbf{r}'(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) &= 0 \\ \text{Odvod integrala s parametrom:} \\ F(x) &= \int_{u(x)}^{v(x)} f(x, y) dy \quad \Rightarrow \\ F'(x) &= \int_{u(x)}^{v(x)} \frac{\partial(f(x, y))}{\partial(x)} dy + f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x) \\ \text{Naravna parameterizacija: } |\mathbf{r}'(s)|^2 &= 1, \quad s = \text{ločna dolžina} \\ \text{Tangentna premica in normalna ravnina: } \vec{e} \text{ ali } \vec{n} &= \dot{\vec{r}}(t_0) \\ \text{Kot med Vektorjema: } \cos \theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \end{aligned}$$

Večkratni integrali

$$\begin{aligned} \text{Dvojni integral: } \iint_D f(x, y) \, dA &= \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx \\ \text{Ploščina: } A &= \iint_D 1 \, dA \\ \text{Prostornina: } V &= \iiint_D f(x, y) \, dA \\ \text{Težišče: } \bar{x} &= \frac{1}{A} \iiint_D x \, dA, \quad \bar{y} = \frac{1}{A} \iiint_D y \, dA \\ \text{Trojni integral: } \iiint_E f(x, y, z) \, dV \\ \text{Masa: } m &= \iiint_E \rho(x, y, z) \, dV \\ \text{Vztrajnostni moment: } I &= \iiint_E r^2 \rho \, dV \\ \text{Povprečna vrednost: } \bar{f} &= \frac{1}{V} \iiint_E f(x, y, z) \, dV \end{aligned}$$

Krivuljni integrali

$$\begin{aligned} \text{S. polje: } \int_C f(x, y, z) ds &= \int_{t_0}^{t_1} f(x(t), y(t), z(t)) * \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt \\ \text{V. polje: } \int_C \mathbf{V} \cdot d\mathbf{r} &= \int_C (Pdx + Qdy + Rdz) = \\ &= \int_{t_0}^{t_1} (P\dot{x} + Q\dot{y} + R\dot{z}) dt \quad d\vec{r} = \dot{\vec{r}} dt = (\dot{x}dt, \dot{y}dt, \dot{z}dt) \\ \text{Neodvisen od poti če } \vec{v} &\text{ potencialno polje (} \textit{rot}\vec{V} = 0 \text{)} \end{aligned}$$

Ploskovni integrali

$$\begin{aligned} \text{S. polje: } \iint_S f \, dS &= \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2} \, du \, dv \\ \text{kjer } E &= \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v \\ z &= z(x, y) \vec{n} = (-z_x, -z_y, 1) \\ \text{V. polje: } \iint_S \mathbf{V} \cdot d\mathbf{S} &= \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v] \, du \, dv \end{aligned}$$

Integralski izreki

$$\begin{aligned} \text{Greenov: } \oint_C (P \, dx + Q \, dy) &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \\ \text{C : zaključena pozitivno orientirana krivulja, ki je rob območja D} \\ \text{Stokesov: } \oint_C \mathbf{V} \cdot d\mathbf{r} &= \iint_S (\nabla \times \mathbf{V}) \cdot d\mathbf{S} \\ \text{C : rob površine S} \\ \text{Gaussov: } \iint_S \mathbf{V} \cdot d\mathbf{S} &= \iiint_E \nabla \cdot \mathbf{V} \, dV \\ \text{S : zaprt površina, ki je rob območja E} \end{aligned}$$

Kompleksna analiza
Elem. komp. fun.

$$\begin{aligned} z &= x + iy, \quad |z| = \sqrt{x^2 + y^2}, \quad \arg z = \arctan \left(\frac{y}{x} \right) \\ e^z &= e^x (\cos y + i \sin y) \\ \log z &= \ln |z| + i(\arg z + 2\pi k) \\ \sin z &= \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2} \\ \sinh z &= \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2} \end{aligned}$$

Analitičnost in odvedljivost in harmoničnost

$$\begin{aligned} \text{C-R pogoj: } \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \text{Če velja je fun. analitična in odvedljiva} \\ \text{Pogoj harmoničnosti: } \nabla^2 u &= \nabla^2 v = 0 / w_{xx} + w_{yy} = 0 \end{aligned}$$

Formule kompleksne analize

$$\begin{aligned} \text{Cauchy's formula: } f(z_0) &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \\ \text{Residue theorem: } \oint_C f(z) dz &= 2\pi i \sum \text{Res}(f, z_k) \\ \textit{Res} f(z) &= \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \\ \text{Laurent series: } f(z) &= \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \end{aligned}$$

Konformne preslikave

$$\begin{aligned} \text{Preslikava: } w &= f(z) = u(x, y) + iv(x, y) \\ \text{Lomljena Linearna preslikava / Mobiusova presl.} \\ f(z) &= \frac{az + b}{cz + d}; ad - bc \neq 0 \end{aligned}$$

Integrali

Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x)$

Po delih: $\int u \, dv = uv - \int v \, du$

Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}}$

Elementarni integrali

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

Odvodi

Pravila odvajanja

$$(f + g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Elementarni odvodi

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

Trigonometrijske funkcije

$$\sin^2 x + \cos^2 x = 1$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$